

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“JNANA SANGAMA”, BELAGAVI - 590018



DEEP LEARNING MINI PROJECT REPORT(CS722T2C)

on

“AI-Based Resume Screening using BERT”

Submitted by

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In partial fulfillment of the requirements for the VII semester

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Under the Guidance of

Ms. Shiji Abraham

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at



SAHYADRI

College of Engineering & Management

An Autonomous Institution

MANGALURU

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EVALUATION RUBRICS

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Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation , Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation , presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
	Total					

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Department of Computer Science & Engineering

DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled “**AI-Based Resume Screening using BERT**” has been carried out by us at Sahyadri College of Engineering & Management, Mangaluru under the supervision of **Ms. Shiji Abraham**, in partial fulfillment of the requirements for the VII semester of **Bachelor of Engineering in Computer Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

The recruitment process involves screening a large number of resumes against specific job requirements, which makes manual evaluation time-consuming and inefficient. Traditional keyword-based resume screening methods may fail to identify suitable candidates when resumes and job descriptions use different words or expressions with similar meanings. This project presents an AI-Based Resume Screening and Job Matching System Using BERT to automate the initial recruitment process using Natural Language Processing and semantic similarity.

The proposed system uses Sentence-BERT (all-MiniLM-L6-v2) to generate 384-dimensional semantic embeddings from resumes and job descriptions. Cosine similarity is then used to measure the relevance between candidate resumes and the selected job description, and candidates are ranked according to their matching scores. The system supports resume files in PDF, DOCX, and TXT formats and provides features such as user authentication, job management, resume uploading, resume parsing, candidate ranking, and screening history. The system is implemented using React.js, FastAPI, Python, and SQLite. The proposed approach helps reduce manual screening effort and provides an automated and systematic method for matching candidates with suitable job requirements.

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I. Introduction

1.1 Overview

The recruitment process is an important activity in every organization. Recruiters are often required to evaluate a large number of resumes for a single job position and identify candidates whose qualifications and skills match the requirements of the organization. Manual resume screening requires considerable time and effort and may result in inconsistent evaluation of candidates.

The proposed project, AI-Based Resume Screening and Job Matching System Using BERT, is designed to automate the initial resume screening and candidate-job matching process. The system uses Natural Language Processing (NLP) and BERT-based semantic representations to compare candidate resumes with job descriptions.

The proposed system uses the Sentence-BERT all-MiniLM-L6-v2 model to generate numerical vector representations, known as embeddings, for resumes and job descriptions. Cosine similarity is then used to determine the semantic similarity between the job requirements and candidate profiles. Candidates are ranked according to their calculated matching scores.

The system is implemented as a full-stack web application using React.js for the frontend and FastAPI for the backend. SQLite is used for persistent data storage, while JWT-based authentication and bcrypt password hashing are used to provide secure access. The system also supports resume processing in PDF, DOCX and TXT formats.

1.2 Scope & Motivation

The motivation behind the proposed system is the increasing difficulty of manually screening large numbers of resumes during the recruitment process. Traditional Applicant Tracking Systems generally rely on keyword-based matching, which may fail to identify suitable candidates when different words or phrases are used to represent similar skills.

The proposed system uses semantic similarity to understand the contextual relationship between resumes and job descriptions. This allows the system to compare the overall meaning of the candidate's resume with the requirements of a particular job rather than relying only on exact keyword matches.

The scope of the project includes automated resume screening, job management, candidate matching, candidate ranking, resume text extraction, screening history and recruiter dashboard functionalities. The system is designed for HR departments and recruitment teams that need to efficiently process large volumes of resumes.

2. Literature Survey

The development of Artificial Intelligence (AI) and Natural Language Processing (NLP) has significantly improved automated recruitment and resume screening systems. Several researchers have explored machine learning, deep learning, Transformer models, semantic similarity, and recommendation techniques for matching resumes with job descriptions.

- [1] **Rojas-Galeano et al.** conducted a bibliometric study on AI-based job-resume matching and highlighted the growing use of NLP, neural networks, and contextual language representations for semantic recruitment.
 - [2] **Devlin et al.** introduced **BERT**, a Transformer-based language model that generates contextual word representations by considering both left and right context, improving various NLP tasks.
 - [3] **Reimers and Gurevych** introduced **Sentence-BERT (SBERT)**, which generates meaningful sentence embeddings that can be efficiently compared using cosine similarity, making it suitable for resume-job matching.
 - [4] **Kaya and Bogers** investigated BERT-based sentence-pair classification for algorithmic recruitment and demonstrated the potential of Transformer models for resume and job-description matching.
 - [5] **Akram et al.** proposed a machine-learning-based resume classification approach to automatically categorize resumes and reduce manual recruitment effort.
 - [6] **Mohanty et al.** developed **Resumate**, an NLP-based resume parsing system that extracts relevant candidate information and improves recruitment efficiency.
 - [7] **Thangaramya et al.** investigated automated resume parsing and ranking using NLP techniques to extract information and rank candidates according to job requirements.
 - [8] **Pradeepa et al.** proposed an intelligent resume evaluation system using machine learning to analyze candidate profiles and support resume assessment.
 - [9] **Jaiswal et al.** developed an NLP-based resume analyzer and job recommendation system for identifying suitable relationships between candidate profiles and job requirements.
 - [10] **Deshmukh and Raut Dahake** compared BERT and Sentence-BERT for automated resume screening and reported improved semantic matching performance using Sentence-BERT.
 - [11] **ResuMatcher** proposed a Transformer-based resume ranking system that uses semantic representations to improve candidate ranking beyond traditional keyword matching.
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- [12] **Liu** proposed a BERT-based deep learning system for intelligent resume-position matching using semantic understanding of resumes and job descriptions.
 - [13] **Kumar and Chellamani** developed a resume screening system using document parsing, Sentence Transformers, embeddings, and cosine similarity for comparing resumes with job requirements.
 - [14] **Tayade et al.** proposed an explainable resume screening framework using Sentence-BERT, skill analysis, section-wise analysis, and weighted ranking for candidate-job matching.
 - [15] **Hussin et al.** developed an NLP-based Applicant Tracking System using resume parsing, Named Entity Recognition, Sentence-BERT, cosine similarity, and TF-IDF for automated screening.
 - [16] **Spoladore et al.** reviewed AI-based recommender systems for jobs and job seekers and identified challenges such as skill representation, dataset bias, interpretability, and fairness.

Overall, the reviewed studies show that BERT and Sentence-BERT-based approaches can capture semantic relationships more effectively than traditional keyword-based methods. Recent studies have explored Large Language Models, zero-shot matching, Transformer networks, learning-to-rank, OCR, and multimodal techniques to improve resume analysis, semantic matching, and candidate recommendation. Based on these findings, the proposed system uses Sentence-BERT embeddings and **cosine** similarity for semantic resume-job matching and candidate ranking.

3. Problem Formulation

3.1 Problem Description

Recruitment teams receive a large number of resumes for different job positions. Manually examining every resume and comparing candidate qualifications with job requirements requires significant time and effort.

Traditional ATS systems mainly depend on keyword matching and may not understand the semantic relationship between different terms. A candidate may have the required skills but describe them using different terminology from the job description.

Therefore, there is a need for an automated system that can understand resume content and job descriptions at a semantic level and provide recruiters with a ranked list of suitable candidates.

3.2 Problem Statement

To develop an AI-based resume screening and job matching system using BERT-based Natural Language Processing that automatically extracts information from candidate resumes, compares resumes with job descriptions using semantic similarity, calculates matching scores and ranks candidates according to their suitability for a particular job.

The system should also provide secure authentication, resume management, job management, screening history and a user-friendly recruiter dashboard..

3.3 Objectives

The main objectives of the proposed system are:

- To develop an automated resume screening and job matching system using BERT-based NLP models to generate semantic embeddings and calculate the similarity between resumes and job descriptions.
 - To rank and manage candidates efficiently based on their matching scores, while supporting resume uploads in PDF, DOCX, and TXT formats.
 - To provide secure user authentication and job management using JWT, allowing recruiters to create, view, and manage job postings and candidates.
 - To maintain screening history and provide a recruiter dashboard for managing resumes, jobs, and previous matching sessions through a responsive and user-friendly web interface.
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3.4 Functional Requirements

- User registration and login.
- JWT-based user authentication.
- Resume upload and text input.
- PDF, DOCX and TXT text extraction.
- Resume creation, viewing and deletion.
- Job creation, viewing and deletion.
- Semantic matching between resumes and job descriptions.
- Candidate ranking based on matching scores.
- Screening history storage.
- Dashboard statistics and candidate search.

3.5 Non-Functional Requirements

- Performance: The system should provide resume processing and matching within a reasonable response time.
 - Security: User credentials should be protected using password hashing and JWT authentication.
 - Usability: The application should provide a simple and user-friendly interface.
 - Reliability: Resume, job and screening information should be stored consistently.
 - Scalability: The system should support an increasing number of resumes and job postings.
 - Compatibility: The application should be accessible using modern web browsers.
 - Responsiveness: The user interface should support desktop, tablet and mobile screens.
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4. PROJECT DESIGN AND IMPLEMENTATION

4.1 Proposed Project Architecture

The proposed system follows a semantic resume screening and job matching architecture. The recruiter first logs into the system and creates or selects a job description. Candidate resumes are uploaded in PDF, DOCX, or TXT format and passed through the resume parsing module to extract the required textual information. The extracted text is then pre-processed and converted into semantic embeddings using the Sentence-BERT model. The same process is applied to the job description.

The generated resume and job-description embeddings are compared using cosine similarity to obtain a matching score. Based on these scores, candidates are ranked according to their relevance to the specified job. The results are then displayed through the recruiter dashboard along with screening history.

4.2 High Level Design

At a high level, the system consists of several major modules: Authentication Module, Job Management Module, Resume Upload and Parsing Module, Text Processing Module, Semantic Matching Module, Candidate Ranking Module, and Recruiter Dashboard.

The Authentication Module manages user registration and login using JWT and bcrypt. The Job Management Module allows recruiters to create and manage job descriptions. The Resume Upload and Parsing Module accepts resumes in supported formats and extracts their textual content. The Text Processing Module prepares the extracted content for semantic analysis.

The Semantic Matching Module uses Sentence-BERT to generate vector embeddings for the resume and job description. Cosine similarity is then calculated between the two embeddings. The Candidate Ranking Module uses the similarity scores to arrange candidates according to their relevance. Finally, the Recruiter Dashboard presents the screening results and maintains the screening history.

4.3 Detailed Design

The system begins with user authentication, where recruiters log in using their registered credentials. After successful authentication, the recruiter can create a job by entering information such as job title, required skills, qualifications, experience, and job description.

Candidate resumes are uploaded through the system in PDF, DOCX, or TXT format. For PDF files,

PyMuPDF is used to extract the textual content, while python-docx is used for DOCX documents. Text files are read directly by the system.

The extracted resume text is pre-processed to remove unnecessary content and prepare the text for semantic analysis. The processed resume and job description are passed to the Sentence-BERT all-MiniLM-L6-v2 model. The model generates a 384-dimensional embedding representing the semantic information contained in the text.

Cosine similarity is calculated between the resume embedding and the job-description embedding. The resulting similarity score represents the semantic relevance of the candidate's resume to the selected job. Candidates are then ranked based on their matching scores, and the results are displayed on the recruiter dashboard.

The screening results and relevant information are stored in the SQLite database, allowing recruiters to access previous screening activities through the screening history module.

4.4 Datasets Description

The proposed system uses resume documents and job descriptions as the primary textual data for semantic matching. Resumes may contain information such as educational qualifications, technical skills, professional experience, projects, certifications, and other relevant details. Job descriptions contain information about the job role, required skills, educational qualifications, experience, and other requirements.

The system supports resumes in PDF, DOCX, and TXT formats. The uploaded documents are parsed and converted into text before being processed by the Sentence-BERT model. The extracted textual information is used to generate embeddings and calculate semantic similarity with the corresponding job description.

For system evaluation, the dataset can consist of multiple resume and job-description pairs representing different job roles and skill requirements. The actual number of resumes, job descriptions, and test cases should be entered based on the dataset used during implementation.

4.5 Tools and Technologies used for Implementation

- Programming Language: Python 3.x
 - Frontend: React.js
 - Backend: FastAPI
 - NLP Model: Sentence-BERT (all-MiniLM-L6-v2)
 - Similarity Measure: Cosine Similarity
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- Database: SQLite
 - PDF Processing: PyMuPDF
 - DOCX Processing: python-docx
 - Authentication: JWT and bcrypt
 - Development Environment: Visual Studio Code
 - Supporting Libraries: NumPy, Pandas, scikit-learn, Sentence Transformers
 - Version Control: Git and GitHub
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5. RESULTS AND DISCUSSION

5.1 Experimentation Environment Set Up

The proposed system was developed and evaluated in a Python-based environment using FastAPI for the backend and React.js for the frontend. Sentence-BERT (all-MiniLM-L6-v2) was used to generate semantic embeddings, and cosine similarity was used to calculate the similarity between resumes and job descriptions. SQLite was used for storing user, job, resume, and screening-related information.

The system was tested using a standard computer system with a suitable processor and sufficient RAM. The required Python libraries, Sentence-BERT model, frontend dependencies, and backend packages were installed and configured before testing.

5.2 Functional Requirements Results Achieved

The developed system was tested for its major functional requirements including user authentication, job creation, resume uploading, resume text extraction, text preprocessing, semantic embedding generation, similarity calculation, and candidate ranking. The system successfully processes resumes in PDF, DOCX, and TXT formats and compares them with the selected job description using Sentence-BERT embeddings and cosine similarity. The generated similarity scores are used to rank candidates according to their relevance.

5.3 Non-Functional Requirements Results Achieved

The system provides a simple and user-friendly interface for recruiters to manage jobs, upload resumes, and view screening results. The use of JWT authentication and bcrypt password hashing provides secure user authentication. The modular FastAPI and React.js architecture supports maintainability, while SQLite provides structured storage for application data. The Sentence-BERT based semantic matching approach enables the system to process resumes and job descriptions based on contextual meaning rather than only exact keyword matching.

5.4 Comparison of Results with Existing Works

Compared with traditional keyword-based resume screening approaches, the proposed Sentence-BERT based system provides semantic matching between resumes and job descriptions. The use of contextual embeddings allows the system to identify relevant information even when different terminology or phrasing is used. The cosine similarity based matching approach provides a simple

method for calculating relevance and ranking candidates. The proposed system combines resume parsing, semantic embedding, similarity calculation, and candidate ranking into a single automated recruitment workflow.

5.5 Project GUI Snapshots

The developed system provides a graphical user interface for user authentication, job management, resume uploading, resume screening, candidate ranking, and screening history. The recruiter can create a job description, upload candidate resumes, and view the generated matching results through the dashboard.

Dashboard Add Resume Add Job Match Candidates

Add Candidate Resume

Upload a document or paste resume content for instant BERT parsing.

Upload Document Paste Text

Candidate Full Name
e.g. Monesh Kumar / Rahul Sharma

Email Address
candidate@example.com

Phone Number
+91 98765 43210

Drag & Drop or Click to Browse
Supports PDF, Word DOCX, and TXT files

Analyze & Save Resume

DashboardAdd ResumeAdd JobMatch Candidates

Add Job Description

Define role parameters, required technical stack, and responsibilities for AI candidate screening.

Job Title *

e.g. DevOps Engineer / Machine Learning Engineer

Company / Organization

e.g. MuSigma / TechCorp Solutions

Required Skills (comma-separated) *

e.g. Python, Docker, Kubernetes, AWS, CI/CD, Linux

Minimum Experience (Years)

0

Full Job Description & Requirements *

Describe the responsibilities, day-to-day work, required qualifications, and technical expectations...

Quick Fill: DevOpsML / AIFull Stack

🚀 Publish Job Description

DashboardAdd ResumeAdd JobMatch Candidates

AI Candidate Matcher

Select a candidate and target job to calculate multi-factor BERT semantic fit scores and skill gaps.

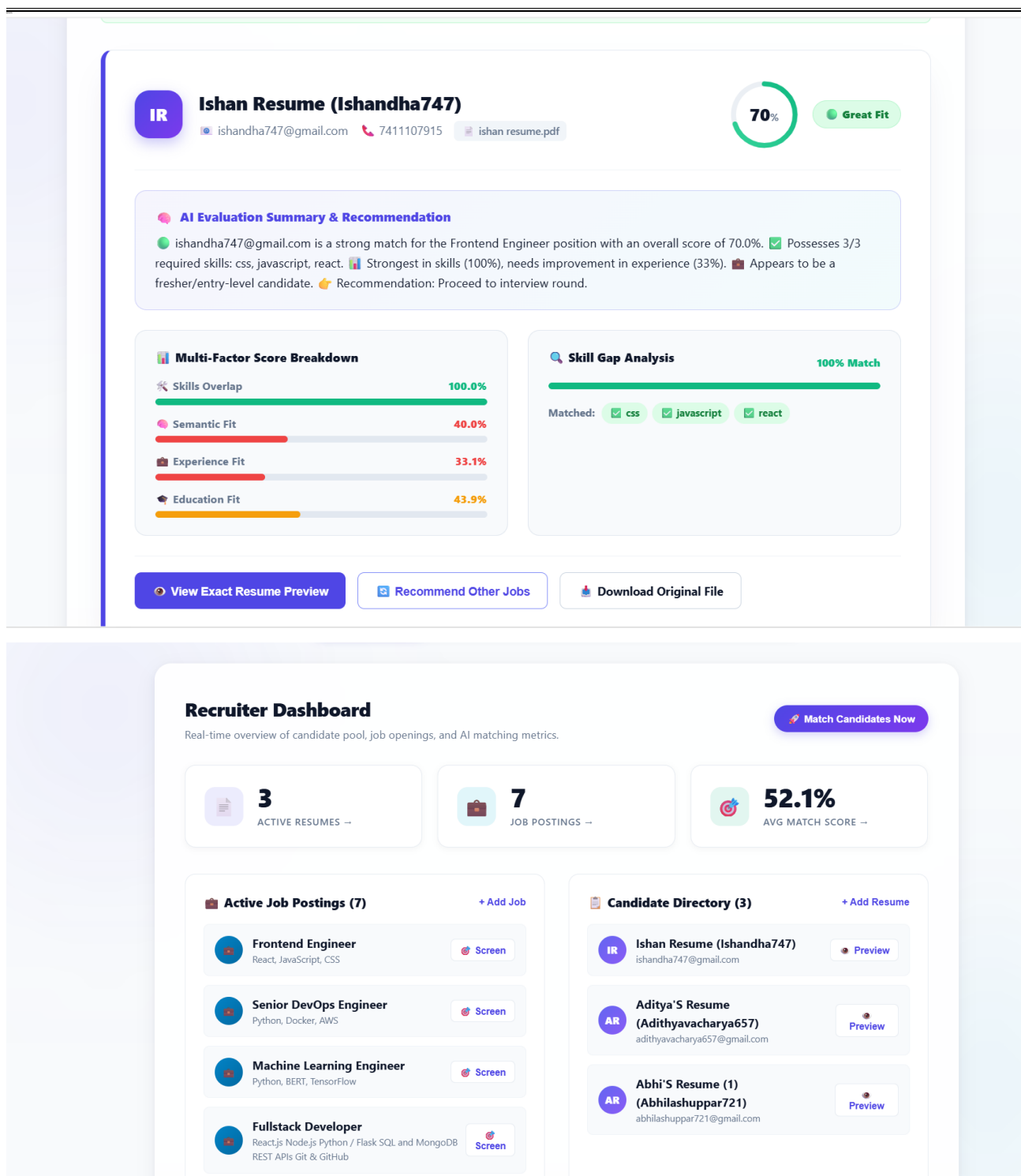
Select Candidate Resume

Ishan Resume (Ishandha747)

Select Target Job

Frontend Engineer

🚀 Calculate Match



5.6 Societal Impact of the Project

The proposed AI-based resume screening and job matching system can reduce the manual effort involved in the initial recruitment process by automatically analysing resumes and matching them with job requirements. The system can help recruiters process a large number of applications in a systematic manner and identify candidates based on the semantic relevance of their resumes to the job description.

The approach can contribute to faster and more organized recruitment while supporting the use of AI-based automation in employment processes.

5.7 SDG Mapped

This project primarily aligns with UN Sustainable Development Goal 8 (Decent Work and Economic Growth) by supporting efficient employment and job-matching processes through AI-based automation. It also contributes to SDG 9 (Industry, Innovation and Infrastructure) through the application of Artificial Intelligence, Natural Language Processing, and Transformer-based models in recruitment systems. The system demonstrates the application of modern AI technologies to improve the efficiency of employment-related processes.

Conclusion and Future Scope

This project successfully designed and implemented an AI-based Resume Screening and Job Matching System using BERT-based Natural Language Processing. The system uses Sentence-BERT (all-MiniLM-L6-v2) to generate semantic embeddings for resumes and job descriptions and cosine similarity to determine the relevance between them. The system supports resume uploading, text extraction, semantic matching, candidate ranking, job management, and screening history. The proposed approach provides an automated solution for initial resume screening and reduces the manual effort required during the recruitment process.

As future scope, the system can be extended by incorporating larger and more diverse resume-job datasets, advanced Transformer and cross-encoder models for improved semantic matching, explainable AI techniques to show why a candidate matches a particular job, and Learning-to-Rank methods for improved candidate ranking. The system can also be extended with job recommendation, skill-gap analysis, multilingual resume processing, and integration with existing Applicant Tracking Systems (ATS) to support larger-scale recruitment applications.

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